

IT221 Network Project

4w4

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Tasks Distribution:

[Nwara algumlas](#) tasks is: Worked in the report.

[Reema Alqarni](#) tasks is: Worked on all network sharing and connectivity setup.

[Shujon Alshehri](#) tasks is: Worked on all network sharing and connectivity setup.

[Fajr Alrubeaan](#) tasks is: Worked on all network sharing and connectivity setup.

[Norah Altamimi](#) tasks is: Worked on all network sharing and connectivity setup.

[Layan Aldosary](#) tasks is: Worked in the report.

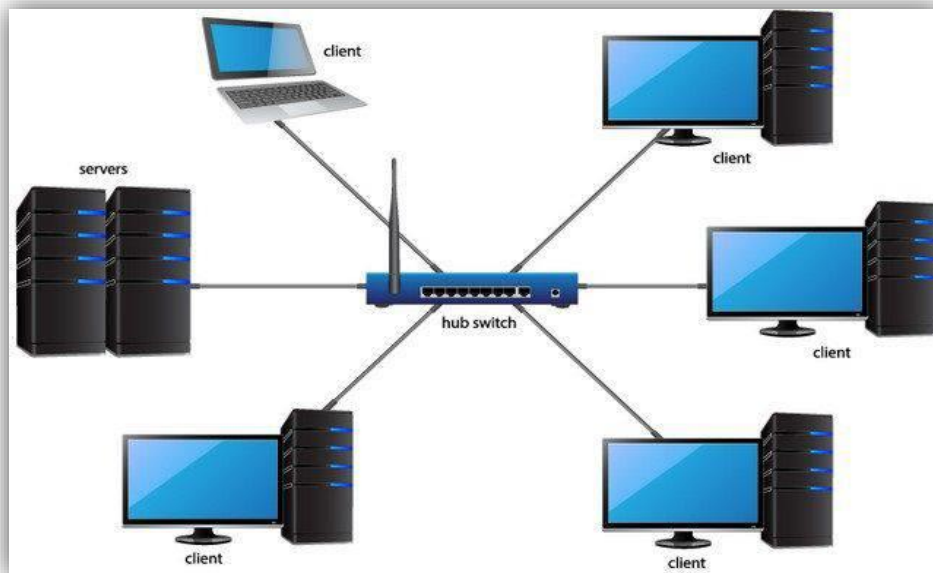
Introduction

In this project, we built a small local network using a router and multiple laptops, following a **Star Topology** where each device connects to a central router. This setup allowed us to observe how devices communicate in a controlled environment without internet access [1].

We assigned static IP addresses to ensure stable connectivity, tested device communication using the ping command, and used Remote Desktop to manage devices remotely [2].

To enhance security, we configured the Windows Firewall to block specific applications and applied simple router settings like MAC address filtering and port security to prevent unauthorized access [3].

Overall, the project improved our understanding of network communication, basic security measures, and teamwork in real-world setups [1], [3].



Tasks

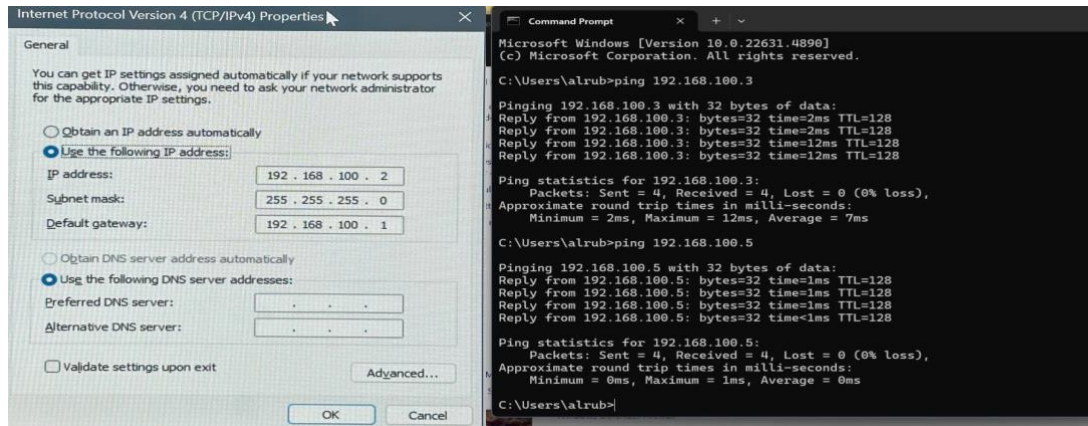
Connect Network:

We started by **turning off Wi-Fi** on all devices to avoid any IP conflicts.

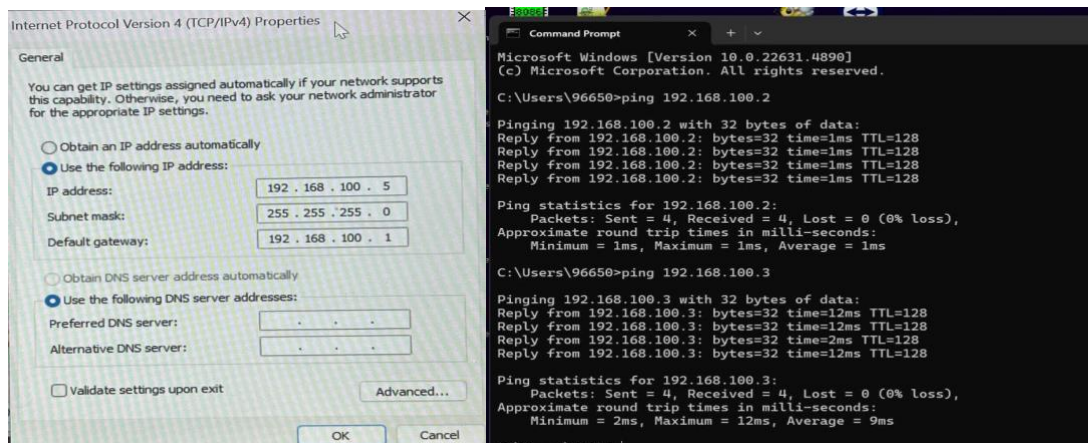
Steps:

1. Connect each laptop to the router using **Ethernet cables**.
2. Opened **Control Panel** → **System and Security** → **Windows Firewall** settings
3. Right clicked the Ethernet adapter → selected **Properties** → then chose **Internet Protocol Version 4 (TCP/IPv4)**.
4. Set a **static IP**:
 - IP Address: 192.168.100.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.100.1
5. Used the **ping command** to check connection (e.g. ping 192.168.100.3)
all devices responded successfully.

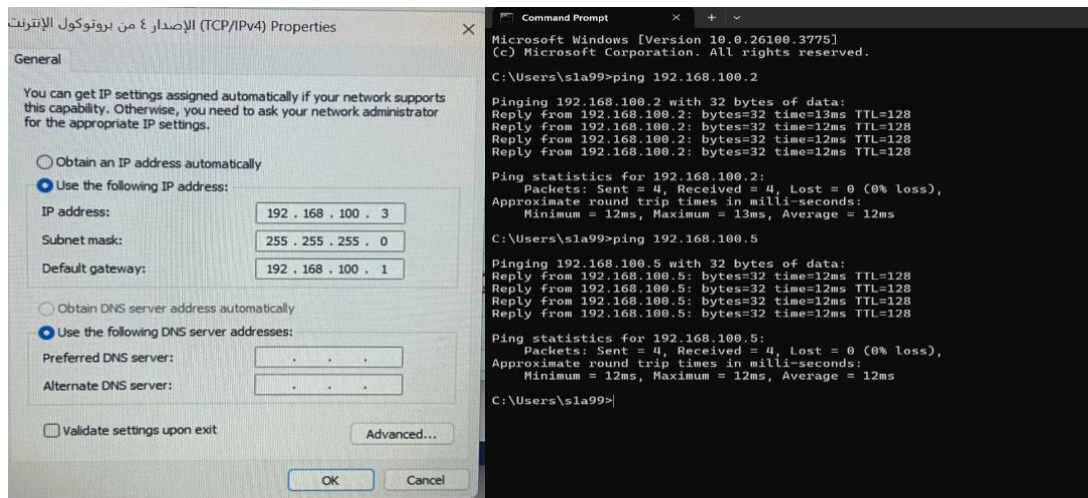
Device 1:



Device 2:



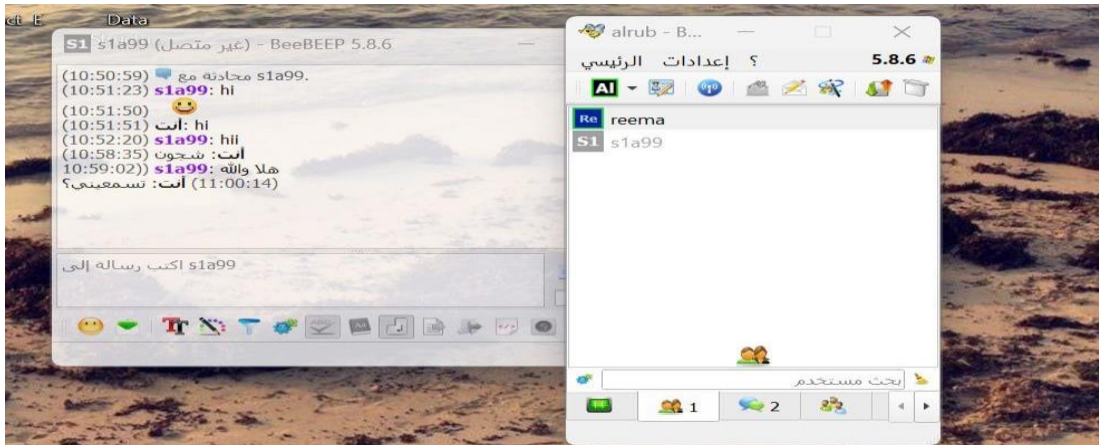
Device 3:



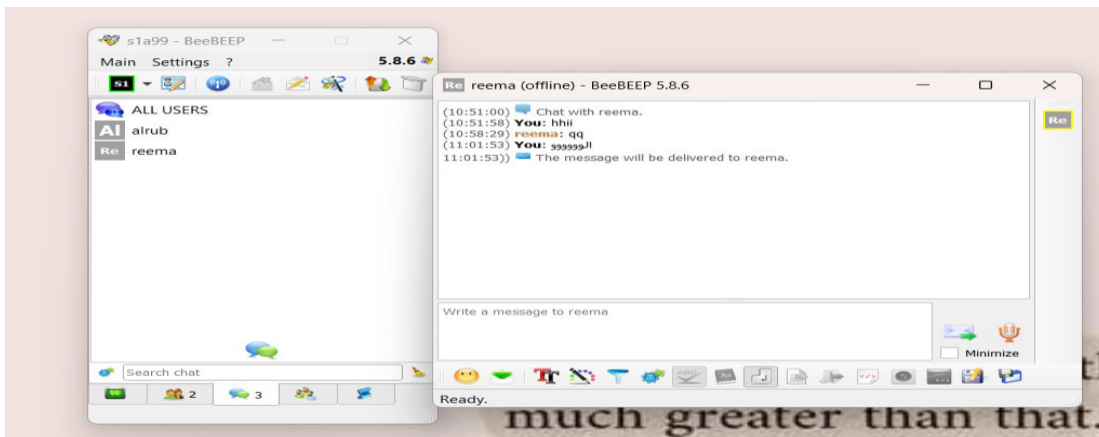
Firewall Block:

We used the BeeBEEP configured the Windows Firewall to block a specific application, such as a chat program, to prevent it from accessing the network or internet.

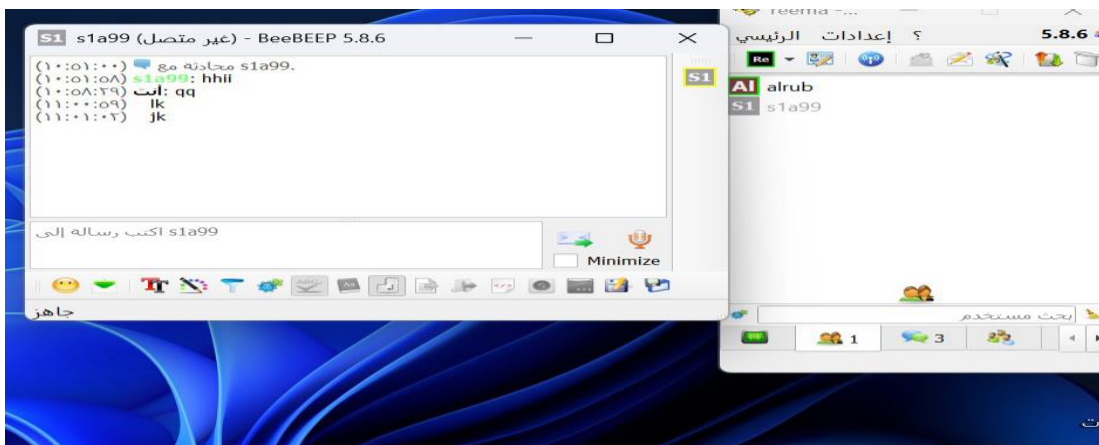
Device 1:



Device 2: (Blocked)



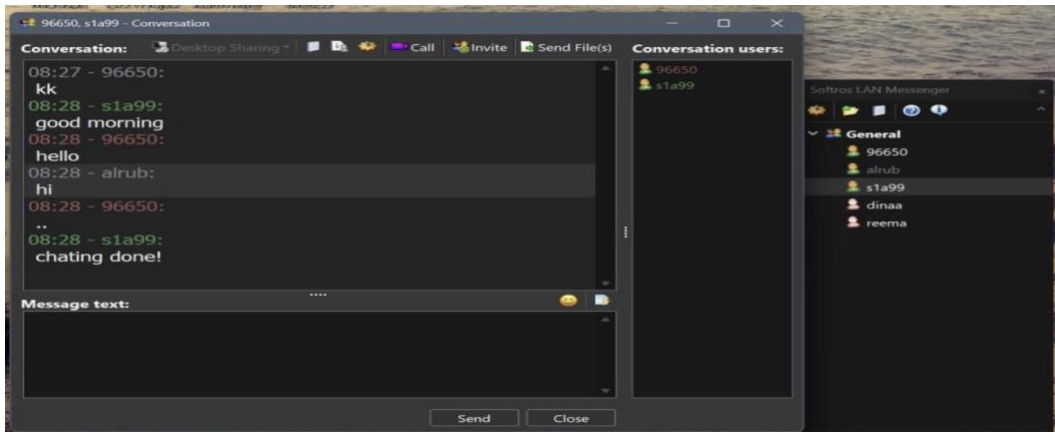
Device 3:



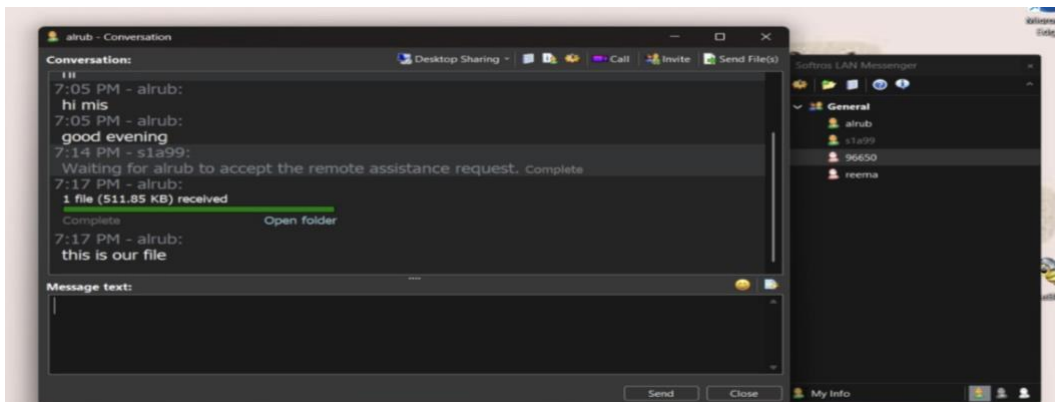
Chatting Program:

We used the LAN Messenger program, which allows communication between devices without needing an internet connection. This software supports sending voice and video files between devices that are connected to the same local network.

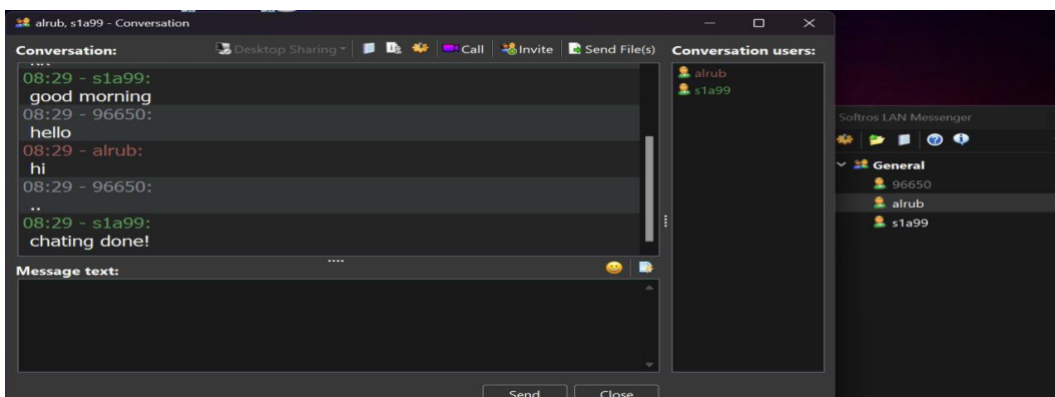
Device 1:



Device 2:



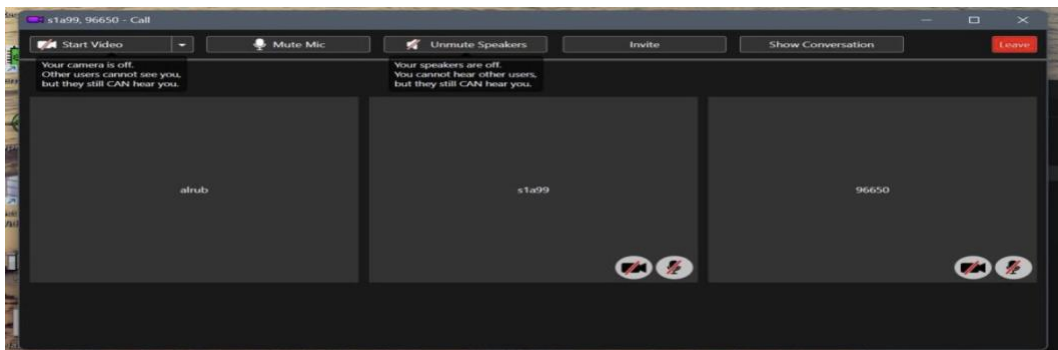
Device 3:



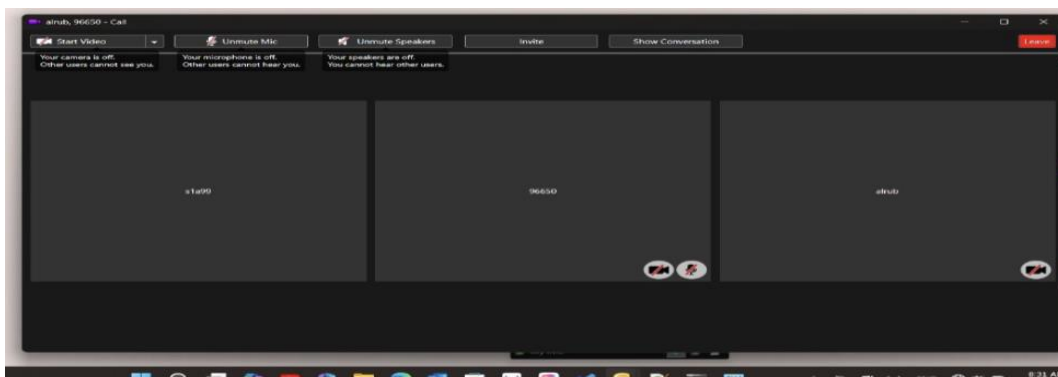
Calling Program:

We used the LAN Messenger communicate between devices without the internet by sending files to any connected device on the network.

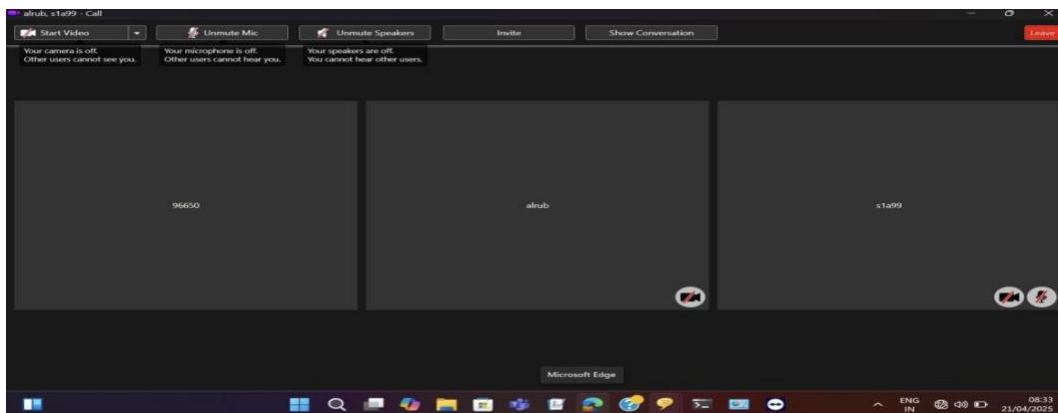
Device 1:



Device 2:



Device 3:



Control Remote Application:

To control another device on the network, we used **TeamViewer**.

Steps we follow:

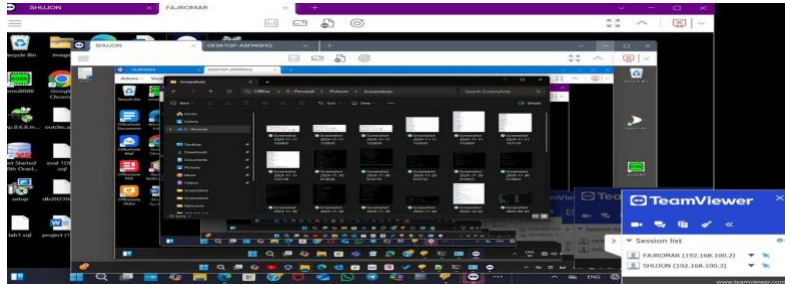
1. TeamViewer installed on both devices.
2. After connecting, a full remote control was successfully established.

This allowed us to control another laptop within the LAN without needing internet. The screenshot shows a successful remote session between two connected devices.

Device 1:



Device 2:



Device 3:



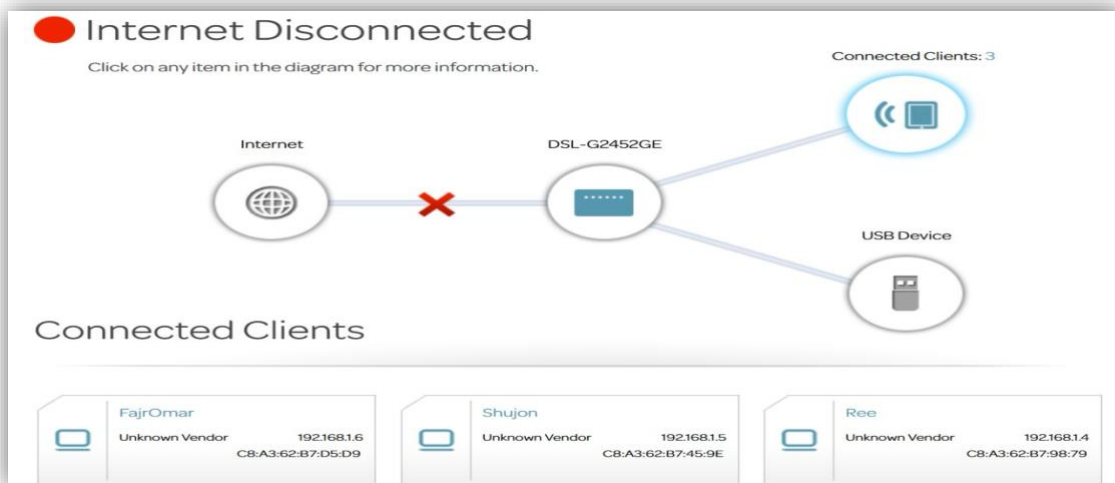
MAC address:

The MAC address (Media Access Control address) is a unique identifier assigned to every network device. It is used within local networks to identify devices, control access, and manage communication between them.

In network management, the blacklist prevents listed devices from connecting to the Wi-Fi, while the whitelist allows only approved devices to access the Wi-Fi network.

We successfully accessed the MAC address page through the router settings. From there, we were able to view all the connected devices on the network and identify which devices were listed in the blacklist.

We also attempted to display the whitelist, which is supposed to show only the allowed devices, but unfortunately, this feature did not work properly during our setup.



Features >> MAC Filter

[IP Filter](#) [WebSite Filter](#)

Enable: ☒

Filtering Mode: ☒ Blacklist ☐ Whitelist

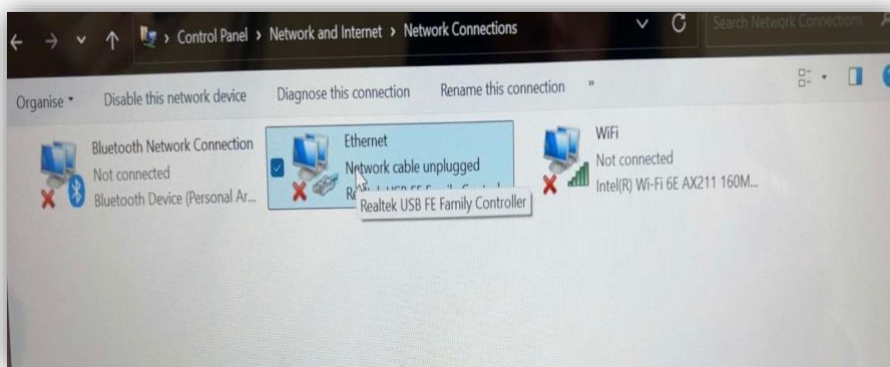
Filter Rule	MAC Address	Enable	Delete
SHO	C8:A3:62:B7:45:9E	Enable	☐

Problems:

While using the ping command to test the network connection between devices, we encountered a “Destination Unreachable” error. This indicated that the devices could not communicate properly over the network.

To troubleshoot the issue, we first suspected a hardware problem, so we replaced the network cables by switching from USB-based connections to more stable Type-C cables.

Even after changing the cables, the error persisted, which led us to check the network equipment. We discovered that the router itself might be the cause of the problem. As a solution, we replaced the old router with a new one, and after the replacement, the connection between the devices was successfully restored and the ping command worked without errors.



```
C:\Users\s1a99>ping 192.168.100.4

Pinging 192.168.100.4 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.100.3: Destination host unreachable.
Reply from 192.168.100.3: Destination host unreachable.

Ping statistics for 192.168.100.4:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\s1a99>
```

References:

- [1] Microsoft, “Change TCP/IP settings,” *Microsoft Support*, 2024. [Online]. Available: <https://support.microsoft.com/en-us/help/15089/windows-change-tcp-ip-settings>
- [2] Microsoft, “How to use Remote Desktop,” *Microsoft Learn*, 2024. [Online]. Available: <https://learn.microsoft.com/en-us/windows-server/remote/remote-desktop-services/welcome-to-rds>
- [3] Cisco, “Firewall Technology Overview,” *Cisco*, 2024. [Online]. Available: <https://www.cisco.com/c/en/us/products/security/firewalls/what-is-a-firewall.html>